Aplica el metodo de la secante para encontrar raices de

1. $f(x) = x^3 - 3x + 2 = 0$

con

$x_1 = -2.6$

$x_2 = -2.4$

$\delta \leq 1.0 \times 10^{-2}$

Solución

$i = 1$

$f(-2.4) = -(2.4)^3 - 3(-2.4) + 2 = -4.624$

$f(-2.6) = -(2.6)^3 - 3(-2.6) + 2 = -7.776$

$$x_3 = -2.4 - \frac{-2.4 + 2.6}{-4.624 + 7.776}(-4.624)$$

$x_3 = -2.106598$

$f(x_3) = -(2.106598)^3 - 3(-2.106598) + 2 = -1.028772$

$i = 2$

$$x_4 = -2.106598 - \frac{-2.106598 + 2.4}{-1.028772 + 4.624}(-1.028772)$$

$x_4 = -2.022641$

$f(x_4) = -(-2.022641)^3 - 3(-2.022641) + 2 = -0.206858$

$i = 3$

$$x_5 = -2.022641 - \frac{-2.022641 + 2.106598}{-0.206858 + 1.028772}(-0.206858)$$

$x_5 = -2.001510$

$f(x_5) = -(2.001510)^3 - 3(-2.001510) + 2 = -0.013611$

$i = 4$

$$x_6 = -2.001510 - \frac{-2.001510 + 2.022641}{-0.013611 + 0.206858}(-0.013611)$$

$x_6 = -2.000021$

$f(x_6) = -(2.000021)^3 - 3(2.000021) + 2 = -1.95 \times 10^{-4} \leq \delta$

Entonces, la raiz de f con $\delta \leq 1 \times 10^{-2}$ es

$x = -2.000021$

[illegible] la secante para encontrar raices de

$2 - f(x) = 8 - 4.5(x - \sin x) = 0$

con

$x_1 = 2$

$x_2 = 3$

$\delta \leq 1 \times 10^{-2}$

Solución

$i = 1$

$f(x_1) = 8 - 4.5(2 - \sin 2) = 3.091838$

$f(x_2) = 8 - 4.5(3 - \sin 3) = -4.864959$

$$x_3 = 3 - \frac{(3-2)(-4.864959)}{-4.864959 - 3.091838}$$

$x_3 = 2.388578$

$f(x_3) = 8 - 4.5(2.388578 - \sin(2.388578)) = 0.328685$

$i = 2$

$$x_4 = 2.388578 - \frac{2.388578 - 3}{0.328685 + 4.864959}(0.328685)$$

$x_4 = 2.427280$

$f(x_4) = 8 - 4.5(2.427280 - \sin 2.427280) = 0.025175$

$i = 3$

$$x_5 = 2.427280 - \frac{2.427280 - 2.388578}{0.025175 - 0.328685}(0.025175)$$

$x_5 = 2.430490$

$f(x_5) = 8 - 4.5(2.430490 - \sin 2.430490) = -1.93 \times 10^{-4} \leq \delta$

Entonces, la raiz de f con $\delta \leq 1 \times 10^{-2}$ es

$x = 2.430490$